Depressive and Bipolar Disorders ✕

Today's goal • 3/3 items

Module 1
Learn With PsycLearn Essentials

Module 2
Opening to Depressive and Bipolar Disorders

Module 3
Depressive and Bipolar Disorders

Module 4
Biological Contributing Factors

Lesson 5: Genetic Contributing Factors

- ✓ Developmental Psychopathology
  Reading • 3 min
- ✓ Genetic Contributing Factors
  Reading • 3 min
- ✓ The Link Between Genes and Psychopathology
  Video • 4 min
- ✓ Practical Scenario
  Real-world application • 5 min
- ✓ Genetic Contributing Factors for Depression
  Reading • 10 min
- ✓ Epigenetic Contributors to Depression
  Video • 3 min
- ✓ Genetic Contributing Factors for Bipolar Disorder
  Reading • 3 min
- ○ Check Your Understanding: Genetic Contributing Factors
  Practice Assignment • 4 min

Lesson 6: Neurobiological Contributing Factors

- ✓ Depressive Disorders: A Case Study
  Reading • 5 min
- ✓ Communication in the Synapse
  Video • 4 min
- ○ **Neurochemical Contributors to Depression**
  Reading • 10 min
- ○ Practical Scenario
  Real-world application • 5 min
- ○ Hormonal Contributors to Depression
  Reading • 5 min
- ○ Brain Regions Linked to Depression
  Video • 4 min
- ○ Neurobiological Contributors to Depressive and Bipolar Disorders
  Reading • 15 min
- ○ Check Your Understanding: Biological Contributing Factors
  Practice Assignment • 6 min

Key Takeaways

- ○ Key Takeaways: Biological Contributing Factors
  Reading • 10 min

Module 4 Quiz

- ○ Module Quiz: Biological Contributing Factors
  Graded Assignment • 9 min

Module 5
Psychological Contributing Factors

Module 6
Treatments for Depressive and Bipolar Disorders

Module 7
Suicide

Module 8
Conclusion

Module 9
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Course Resources

Module 11
PsycLearn Essentials APA Student Resources

Neurochemical Contributors to Depression

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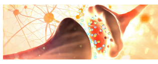

Neurochemical Factors

Neurotransmitters are chemicals that are released by neurons to transmit messages across the junctions or synapses between neurons. Neurotransmitters can serve either an excitatory or inhibitory function. Excitatory neurotransmitters promote the conduction of signals across a nerve synapse and inhibitory neurotransmitters hinder or block the conduction of signals across a nerve synapse. The primary neurotransmitters associated with depression are serotonin, noradrenaline (also called norepinephrine), glutamate, and gamma-aminobutyric-acid, commonly called GABA (Luscher et al., 2011; Werner & Covellas, 2010).

- **Serotonin.** Serotonin is an excitatory neurotransmitter involved in many psychological processes, including the regulation of mood, emotional processing, appetite, sexual desire and performance, pain processing, hallucinations, reflex regulation, and sleep. Low levels of serotonin are thought to be among the primary causes of depression. Many of the medications used to treat depression, such as selective serotonin reuptake inhibitors (SSRIs), focus on raising serotonin levels in the brain.
- **Noradrenaline.** Noradrenaline, also known as norepinephrine, is an excitatory neurotransmitter involved in the regulation of mood and other psychological processes. Some medications for depression focus on raising both serotonin and noradrenaline levels, such as serotonin norepinephrine reuptake inhibitors (SNRIs). SNRIs are a group of antidepressants that inhibit the reabsorption of both serotonin and noradrenaline by the presynaptic neurons.
- **Glutamate.** Glutamate is the most common excitatory neurotransmitter in the brain and is important for memory, motor, and mood regulation. Because of its broad effect throughout the nervous system, glutamate is thought to play at least some role in many disorders, including depressive disorders.
- **GABA.** GABA, or gamma-aminobutyric acid, is an inhibitory neurotransmitter. GABA plays a prominent role in the control of stress; when GABA levels are low, as they are in individuals with depression, stress levels are higher (Luscher et al., 2011). GABA acts to relax the brain, making us feel good. Stress is a major risk factor for depression. GABA and glutamate have opposite actions in the brain, so they typically exist in a balance. Because of their counteracting effect, an imbalance between GABA and glutamate can have dramatic effects on mood and anxiety. Benzodiazepines, a major category of antianxiety medications, enhance the activity of GABA.

When the levels of serotonin, noradrenaline, and glutamate neurotransmitters are low, as they are in individuals with depression, mood is low. The major psychotropic medications for depression are designed to increase levels of these neurotransmitters in the brain, thereby increasing mood. GABA, by contrast, is an inhibitory neurotransmitter, whose effect is to calm brain activity. Individuals with depression have been shown to have low levels of GABA, which is why some treatments for depression attempt to increase the levels of this neurotransmitter so as to increase its calming effect on the brain (Petty, 1995; Prévot & Sibille, 2021).

It may seem like treating depression is nothing more than raising or lowering levels of neurotransmitters, but this is not the case. There are clear benefits to increasing mood and calming brain activity but much more is involved in treating depression, which you will learn about in the later sections of this module.

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